# Intelligent Academic Operations & Attendance Management Platform

25% Project Implementation Progress and Design


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**Guide**

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# INTRODUCTION

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- Academic institutions manage large amounts of **student, timetable, attendance, and academic information**.
- Many activities still depend on **manual records, spreadsheets, and disconnected systems**.
- Changes such as **holidays, class swaps, substitutions, cancellations, and rescheduling** make academic management more complex.
- Students and teachers need a **centralized platform** to access accurate and up-to-date academic information.
- Attendance data can provide valuable insights beyond simply recording **Present/Absent**.
- SecureTrack X proposes a **secure and intelligent platform** that connects academic operations, attendance management, analytics, and user services.

# PROBLEM STATEMENT

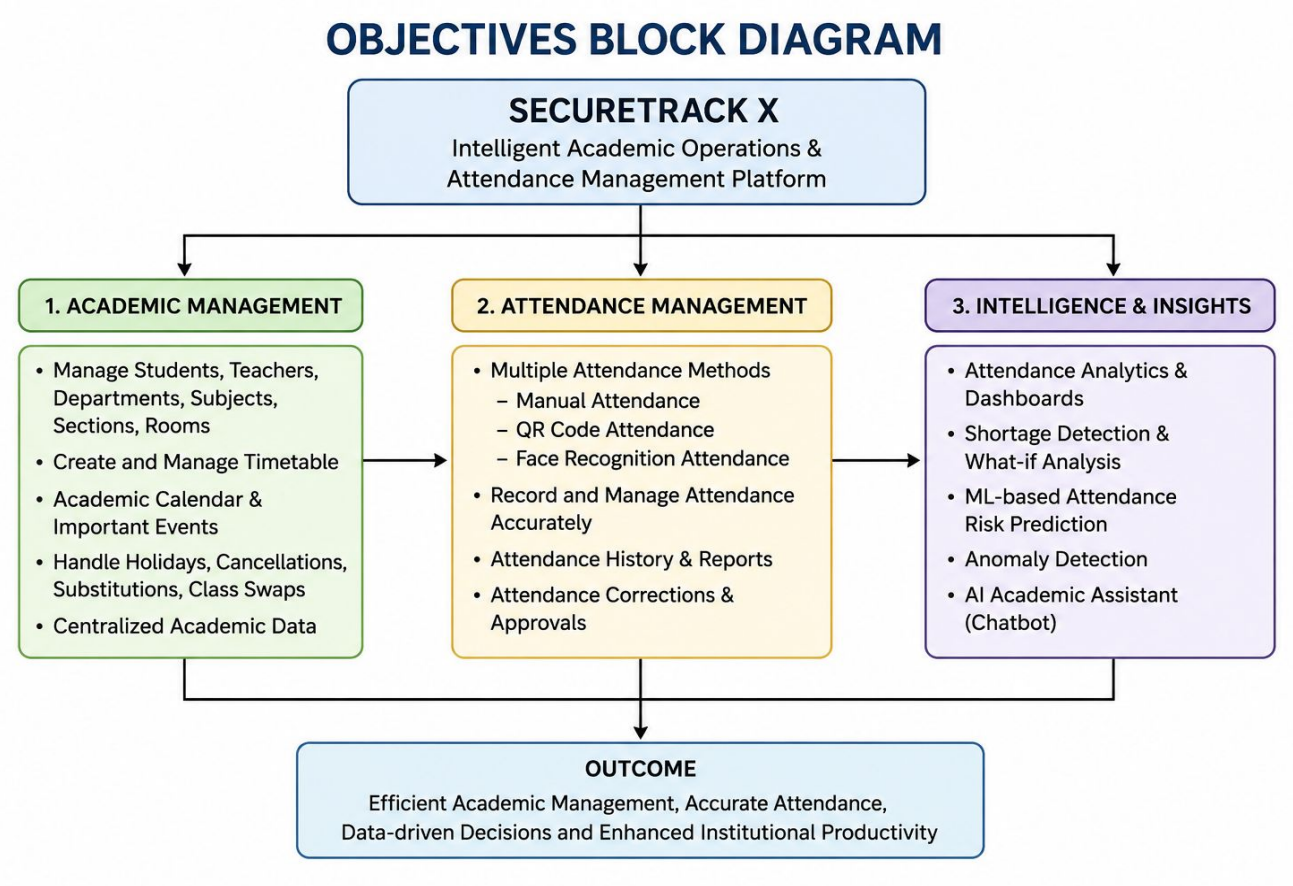
Academic institutions face challenges in managing dynamic academic activities and attendance:

- Attendance is often maintained through **manual records or disconnected systems**.
- **Timetable changes** are difficult to track consistently.
- Unexpected events such as **holidays, class cancellations, substitutions, and rescheduling** can create inconsistencies.
- Students may not have a **centralized view** of attendance, timetable, notices, and academic updates.
- Teachers and administrators spend time on **manual attendance management and report preparation**.
- Traditional systems provide limited **attendance analytics and early identification of shortages**.
- Lack of proper access control and tracking can make sensitive academic data **difficult to manage securely**.

**Key Problem: There is a need for a centralized system that can maintain the planned academic schedule, record actual academic activities, securely manage attendance, and provide meaningful insights from the collected data.**

# OBJECTIVES

- Centralize **students, teachers, subjects, sections and academic schedules**.
- Manage dynamic academic events such as **holidays, cancellations, substitutions and class swaps**.
- Support multiple attendance methods including **manual, QR and face recognition**.
- Provide **role-based portals** for students, teachers and administrators.
- Provide **attendance analytics, shortage detection and what-if analysis**.
- Develop **ML-based attendance risk prediction and anomaly detection**.
- Provide secure **notifications, reports and audit trails**.
- Integrate an **AI academic assistant** for authorized academic information and queries



# EXISTING SYSTEM & LIMITATIONS

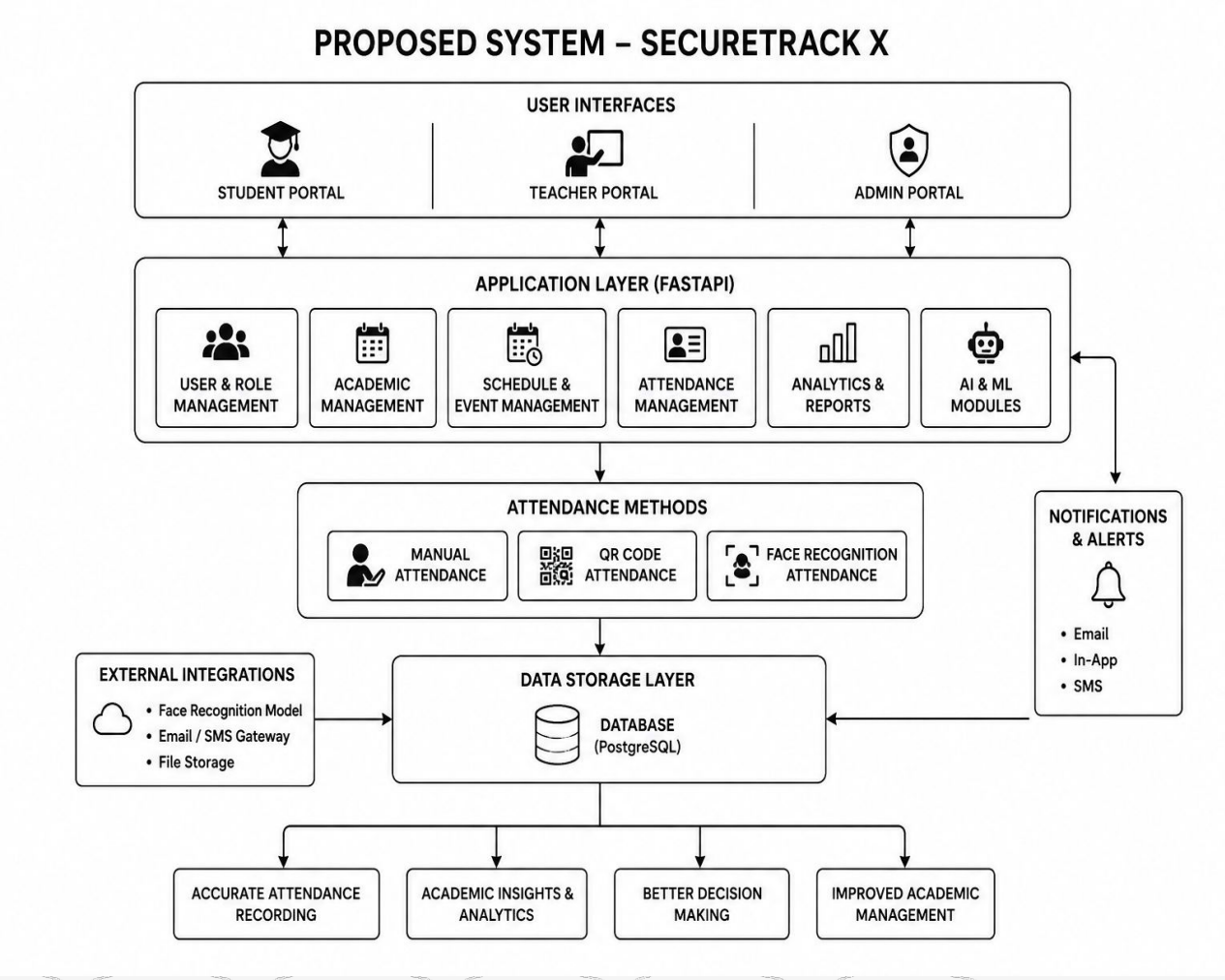
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- Attendance is commonly managed through **manual registers, spreadsheets or basic digital systems**.
- Academic information such as **timetable, calendar and attendance** may be handled separately.
- Timetable-based systems often assume a **fixed schedule**.
- Real-world changes such as **holidays, cancellations, substitutions and class swaps** can make planned schedules inaccurate.
- Attendance records may require **manual correction and verification**.
- Limited analytics make it difficult to identify **attendance trends and shortage risks**.
- Students and teachers may have limited access to **personalized academic insights**.

**KEY GAP** : Existing systems primarily record planned academic activities and attendance, but may not effectively represent dynamic academic changes and provide intelligent insights.

# PROPOSED SYSTEM

- **Centralized academic management** for students, teachers, subjects, sections and schedules.
- **Dynamic academic scheduling** that handles holidays, cancellations, substitutions, class swaps and rescheduling.
- **Multi-modal attendance** using manual, QR and face-recognition methods.
- Separate **Student, Teacher and Admin portals** with role-based access.
- **Attendance analytics** for trends, shortages and student monitoring.
- **Intelligent features** for attendance risk prediction, anomaly detection and what-if analysis.
- **Notifications and reports** for timely academic communication.




- **Secure and auditable operations** using authentication, authorization and activity tracking.

# USER ROLES & PERMISSIONS

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SecureTrack X provides separate access levels based on the user's role.

| Admin | Teacher | Student |
|---|---|---|
| Manage users | Manage assigned classes | View own attendance |
| Manage departments & subjects | Mark attendance | View timetable |
| Manage timetable & calendar | View student attendance | View academic calendar |
| Manage academic events | Review corrections | Apply for leave |
| View institution-wide analytics | Approve/review requests | Request attendance correction |
| Manage notices | Generate class reports | View notices & notifications |
| View audit logs | Request schedule changes | Use AI academic assistant |

# MODULE DESIGN

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SecureTrack X is divided into the following major modules:

- **User & Role Management**
  - Admin, Teacher and Student accounts
  - Role-based permissions
- **Academic Management**
  - Departments, subjects, sections and students
  - Academic year and semester
- **Schedule & Event Management**
  - Timetable and academic calendar
  - Holidays, cancellations, substitutions, swaps and rescheduling
- **Attendance Management**
  - Manual attendance
  - QR-based attendance
  - Face-recognition attendance
  - Attendance history and corrections
- **Analytics & Reporting**
  - Attendance trends
  - Shortage detection
  - Dashboards and reports
  - What-if analysis
- **AI & Intelligence**
  - Attendance risk prediction
  - Anomaly detection
  - AI academic assistant
- **Notification & Audit**
  - Notices and alerts
  - Activity tracking and audit logs

# USE CASE DIAGRAM

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**Admin**

- ➔ Manage users and academic data
- ➔ Manage timetable and academic events
- ➔ Manage notices
- ➔ View analytics and reports
- ➔ Monitor system activity

**Teacher**

- ➔ View assigned classes
- ➔ Mark attendance
- ➔ Manage attendance corrections
- ➔ View student attendance
- ➔ Generate class reports
- ➔ Request schedule changes

**Student**

- ➔ View personal attendance
- ➔ View timetable and academic calendar
- ➔ Apply for leave
- ➔ Request attendance correction
- ➔ View notices and notifications
- ➔ Access academic AI assistant

# SYSTEM WORKFLOW

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User Login

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Role Identification

↓

Access Relevant Portal

↓

View Academic Schedule

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Check Actual Class Session

↓

Attendance Session

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Manual / QR / Face Recognition

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Attendance Validation

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Store Attendance

↓

Analytics & Reports

↓

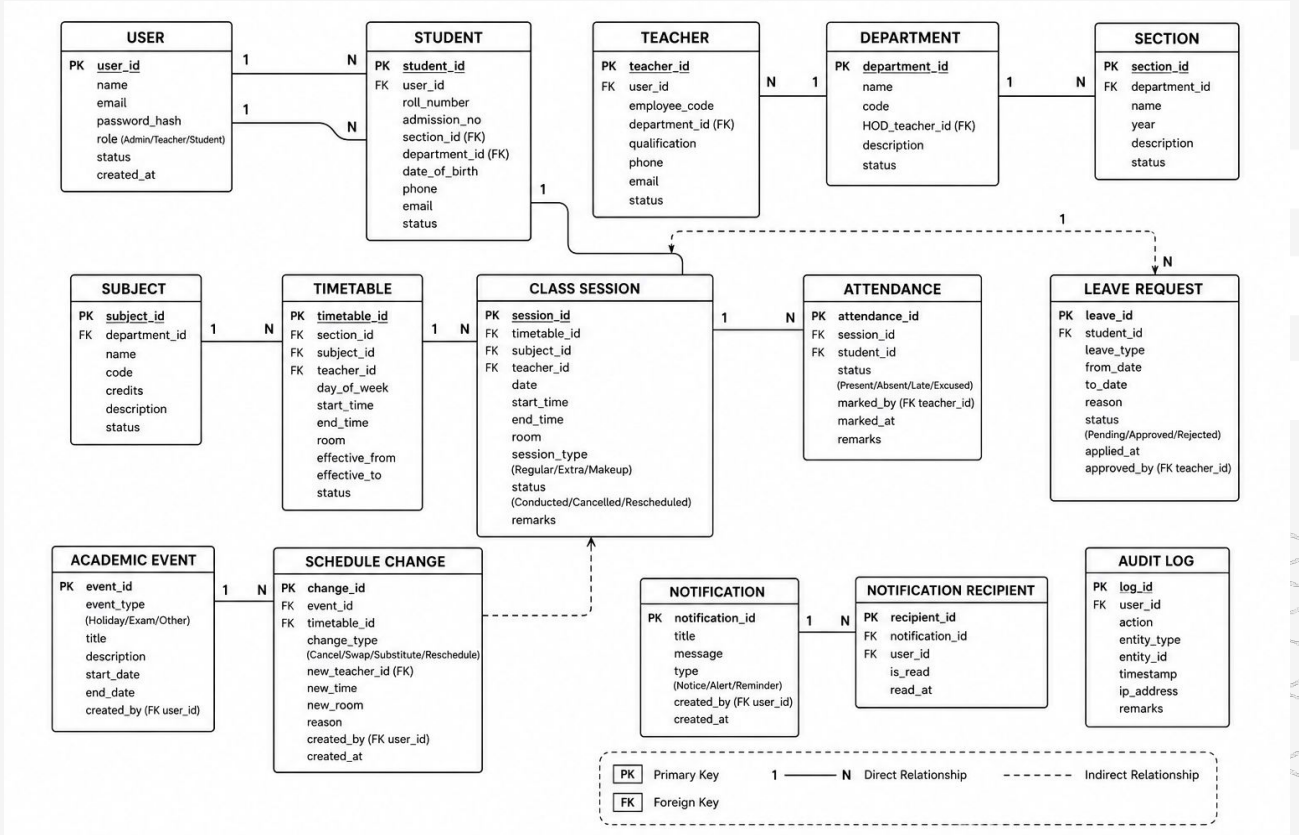
Notifications / Insights

# ER DIAGRAM / DATABASE DESIGN

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## Database Design

The system uses a **relational database** to maintain structured academic, user and attendance information.




## Core Entities

- **User** — authentication and role information
- **Student** — student-specific academic details
- **Teacher** — faculty information
- **Department** — academic department
- **Section** — student grouping
- **Subject** — academic subjects
- **Timetable** — planned class schedule
- **Academic Event** — holidays, exams and special events
- **Class Session** — actual class conducted
- **Attendance** — student attendance for a session
- **Leave Request** — student leave applications
- **Attendance Correction** — attendance correction requests
- **Notification** — system alerts and announcements
- **Audit Log** — record of important system actions

# DATABASE DESIGN & DATA FLOW

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## Database Design

The relational database is designed to maintain the connection between **academic planning, actual class sessions and attendance records**.

## Supporting Data

- **Academic Events** → holidays, exams and special events
- **Schedule Changes** → cancellation, swap, substitution and rescheduling
- **Leave Requests** → student leave workflow
- **Notifications** → alerts and announcements
- **Audit Logs** → track important system activities

## Key Design Principle

"Attendance is linked to an actual Class Session, not directly to a static timetable entry"

This allows the system to correctly handle:

**Holiday → Cancellation → Swap → Substitution → Rescheduling → Compensatory Class**

## Core Data Flow

Academic Structure

↓

Department → Section → Student

↓

Subject + Teacher

↓

Timetable

↓

Class Session

↓

Attendance

↓

Analytics & Reports

# BACKEND / API DESIGN

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**Backend & API Design**

SecureTrack X uses a **RESTful backend architecture** to connect the user portals with the database and application services.

**Backend Components**

- **FastAPI** — REST API framework
- **JWT Authentication** — secure user authentication
- **Role-Based Access Control (RBAC)** — permission enforcement
- **Service Layer** — application/business logic
- **Repository/Data Access Layer** — database operations
- **SQLAlchemy ORM** — database interaction
- **PostgreSQL** — persistent data storage
- **API Validation & Error Handling** — reliable request processing

# BACKEND / API DESIGN

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## User Interface Design

SecureTrack X provides **role-specific dashboards** designed around the needs of each user.

### Student Portal

- Personal attendance dashboard
- Subject-wise attendance
- Timetable & academic calendar
- Notices & notifications
- Leave and correction requests
- Attendance insights
- AI academic assistant

### Teacher Portal

- Today's classes
- Assigned students & subjects
- Attendance management
- Class reports & analytics
- Leave/correction requests
- Schedule-change requests
- Notices & notifications

### Admin Portal

- Institution-wide dashboard
- User & academic management
- Timetable & calendar management
- Schedule changes
- Attendance analytics
- Reports, notices and audit logs

# TECHNOLOGY STACK

| Layer | Technology |
|---|---|
| **Frontend** | React / TypeScript |
| **Backend** | FastAPI / Python |
| **API** | RESTful APIs |
| **Database** | PostgreSQL |
| **ORM** | SQLAlchemy |
| **Authentication** | JWT |
| **AI / Face Recognition** | FaceNet + TensorFlow Lite |
| **Containerization** | Docker |
| **Version Control** | Git / GitHub |
| **API Testing** | Swagger / Postman |

## Development Tools

- Visual Studio Code
- PostgreSQL / pgAdmin
- Docker
- GitHub
- Postman

# SECURITY DESIGN

SecureTrack X is designed with security at multiple levels:

- **JWT-based authentication** for secure user login and API access.
- **Role-Based Access Control (RBAC)** for Admin, Teacher and Student permissions.
- **Object-level authorization** to prevent users from accessing unauthorized records.
- **Password hashing** to avoid storing plain-text passwords.
- **Input validation** to prevent invalid or malicious requests.
- **API access control** for protected endpoints.
- **Attendance validation** to prevent duplicate or invalid attendance records.
- **Audit logging** to track important changes and administrative actions.
- **Secure configuration** using environment variables for sensitive credentials.

**Security Principle**

**Authenticate → Authorize → Validate → Record → Audit**

# 25% IMPLEMENTATION PROGRESS

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**Completed**

- Project structure and layered backend architecture
- PostgreSQL database setup
- SQLAlchemy models and relationships
- User registration and login
- JWT-based authentication
- Role-based access control
- Student management APIs
- Class/subject management APIs
- Attendance marking and retrieval
- Duplicate attendance prevention
- Attendance percentage calculation
- Basic attendance dashboard
- CSV attendance reporting
- Docker-based deployment

**Next Implementation Phase**

- Timetable & academic calendar
- Class-session management
- Dynamic schedule changes
- QR attendance
- Face-recognition integration
- Advanced analytics
- Attendance risk prediction
- Notifications
- Audit logging
- AI academic assistant

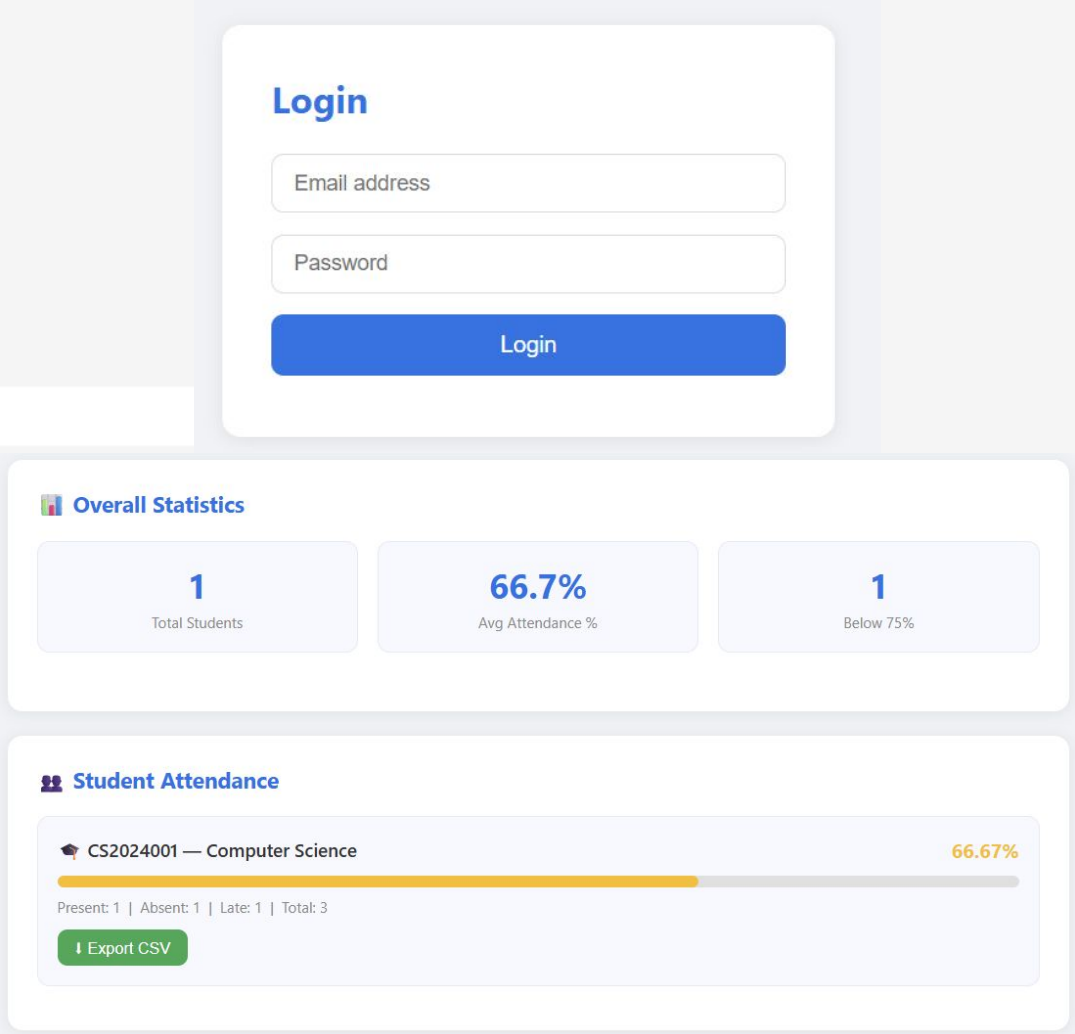
# CURRENT IMPLEMENTATION & WORKING DEMO

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## Current Implementation

The implemented foundation currently demonstrates:

- **User registration and login**
- **JWT-based authentication**
- **Role-based access control**
- **Student and class management**
- **Attendance marking**
- **Duplicate attendance prevention**
- **Attendance retrieval and summary**
- **Attendance percentage calculation**
- **Basic dashboard**
- **CSV report generation**
- **REST API documentation through Swagger**
- **Dockerized deployment**



# CHALLENGES & MITIGATION

| Challenge | Proposed Approach |
|---|---|
| Dynamic timetable changes | Separate **planned timetable** from **actual class sessions** |
| Emergency holidays / cancellations | Academic event and session status management |
| Class swaps / teacher substitutions | Schedule-change workflow with approval |
| Duplicate attendance | Database constraints + backend validation |
| Unauthorized data access | JWT + RBAC + object-level authorization |
| Attendance correction disputes | Correction request + approval + audit trail |
| Multiple attendance methods | Common attendance API for manual, QR and face recognition |
| Large attendance data | Structured relational database + optimized queries |

**Key Design Principle:** The system should record what actually happened, not blindly assume that the original timetable was followed.

# FUTURE DEVELOPMENT & IMPLEMENTATION ROADMAP

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**Phase 2 — Academic Operations**

- Timetable management
- Academic calendar
- Class-session management
- Holiday and cancellation handling
- Class swaps and teacher substitutions
- Rescheduling and compensatory classes

**Phase 3 — Smart Attendance**

- Dynamic QR attendance
- Integration of the existing **Face Recognition module**
- Attendance correction workflow
- Leave management
- Advanced attendance reports

**Phase 4 — Intelligence**

- Attendance trend analysis
- Shortage prediction
- **ML-based attendance risk prediction**
- Attendance anomaly detection
- What-if attendance analysis

**Phase 5 — AI & Production**

- RAG-based academic assistant
- Intelligent notifications
- Audit and monitoring
- Automated testing
- CI/CD and performance optimization

# CONCLUSION

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- **SecureTrack X** proposes a centralized platform for academic operations and attendance management.
- The system connects **academic scheduling, actual class sessions and attendance records**.
- It supports role-specific access for **Admin, Teacher and Student**.
- The design considers real-world academic changes such as **holidays, cancellations, substitutions, class swaps and rescheduling**.
- Multiple attendance methods such as **manual, QR and face recognition** are planned.
- Attendance data will be used for **analytics, prediction and intelligent decision-making**.
- Security is incorporated through **authentication, authorization and auditability**.
- The current implementation establishes the **core backend, database, authentication, RBAC and attendance foundation**.

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# THANK YOU




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